

Chapter 3 – Correlation

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1 Introduction

1. Unlike simple linear regression, we only have observations with one set of variables in time series data.
2. All time series models assume that current observations must be correlated somehow with observations in previous periods in some sort of regular manner.
3. Hence once the trend and seasonality is removed from the data, we look at how the observations are correlated.
4. We introduce some basic notation for this course.
5. The mean of a random variable x is $\mu_x = E[x]$.
6. The variance of a random variable x is $\sigma_x^2 = E[(x - \mu_x)^2]$.
7. If there are two random variables x and y , then the means are μ_x and μ_y respectively.
8. The covariance of x and y is denoted by $\text{Cov}(x, y) = E[(x - \mu_x)(y - \mu_y)]$, that is, it is the expected cross-deviation of each variable from their respective means.
9. If we have sample observations of x and y , say, $(x_1, \dots, x_n)$ and $(y_1, \dots, y_n)$, then the sample means are $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ and $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$; sample variances are $s_x^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$ and $s_y^2 = \frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n-1}$ and the sample covariance is $\gamma_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n-1}$.

2 Stationarity and Ergodicity

1. With time series data, we have a whole ensemble of serial observations $(x_1, x_2, \dots, x_n)$.
2. The mean of the ensemble over the t th period is $\mu(t) = E[x_t]$.
3. If there are trends or seasonality then $\mu(t)$ will be a function of time and it fluctuates.
4. If the theoretical value of $\mu(t)$ is a constant μ , then we say that the series is *stationary*.
5. In this case, if we have n observations from the series, then we can estimate μ from a sample mean of the n observations $\bar{x} = \frac{1}{n} \sum_{t=1}^n x_t$.
6. Sometimes as the length of the time series increases, the ensemble mean converges to some constant mean value μ . If this happens, we say that the series is *ergodic* and in mathematical notation, we have

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{t=1}^n x_t = \mu.$$

7. The variance function of a time series with stationary mean is

$$\sigma^2(t) = E[(x_t - \mu)^2].$$

8. If the model is also assume to be stationary in the variance, then an estimate of the variance function is given by the sample variance function

$$s^2 = \frac{\sum_{i=1}^n (x_t - \bar{x})^2}{n - 1}.$$

3 Autocorrelation

1. The mean and variance play an important role in the study of statistical distributions because they summarise two key properties of the distribution – a central value and the spread.
2. Similarly in time series models, key roles are not only played by the mean and variance, but also by another property which looks at the serial correlation.
3. We are interested to see if preceding observations of the time series are correlated to the current observation as well as any future observations.
4. Consider a time series that is stationary in the mean and variance. However, the observations may be correlated and the model is said to be *second-order stationary* if the correlation between the observations depends only on the number of time steps separating them.
5. This number of time steps is known as the *lag*.
6. A correlation of a variable with itself at different times is known as *autocorrelation*. Some authors also call it *serial correlation*.
7. If a time series model is second-order stationary, then the *autocovariance function* (acvf) is

$$\gamma_k = E[(x_t - \mu)(x_{t+k} - \mu)].$$

8. The function does not depend on t because since the time series is stationary, the mean μ is constant over time and thus both x_t and x_{t+k} have the same mean μ .
9. The lag k *autocorrelation function* is defined by

$$\rho_k = \frac{\gamma_k}{\sigma^2},$$

where σ^2 is the variance of x_t (and also of x_{t+k}).

10. Given the times series data, the acvf and the acf can be estimated by the sample equivalents. In fact, this is how **R** will calculate them.
11. The sample acvf c_k is calculated by

$$c_k = \frac{1}{n} \sum_{t=1}^{n-k} (x_t - \bar{x})(x_{t+k} - \bar{x}).$$

12. Note that the sample acvf at lag 0 is simply the sample variance calculated with denominator n instead of the usual $n - 1$.
13. If we need to calculate c_k up to lag k , we need to have more than k terms in the series.
14. The sample acf is given by

$$r_k = \frac{c_k}{c_0} = \frac{\sum_{t=1}^{n-k} (x_t - \bar{x})(x_{t+k} - \bar{x})}{\sum_{t=1}^n (x_t - \bar{x})^2}.$$

15. **Example 1.** Use the file wave.txt found in the course website. Note that you need to save it somewhere and then use **R** to read it in as an **R** data set. The data set consists of wave height readings (mm relative to still water level) measured at the centre of an experimental wave tank that is programmed to simulate waves in a rough sea with no discernible patterns or trends. The sampling interval is 0.1 second and the record length is 39.7 seconds. We need to read the file in first.

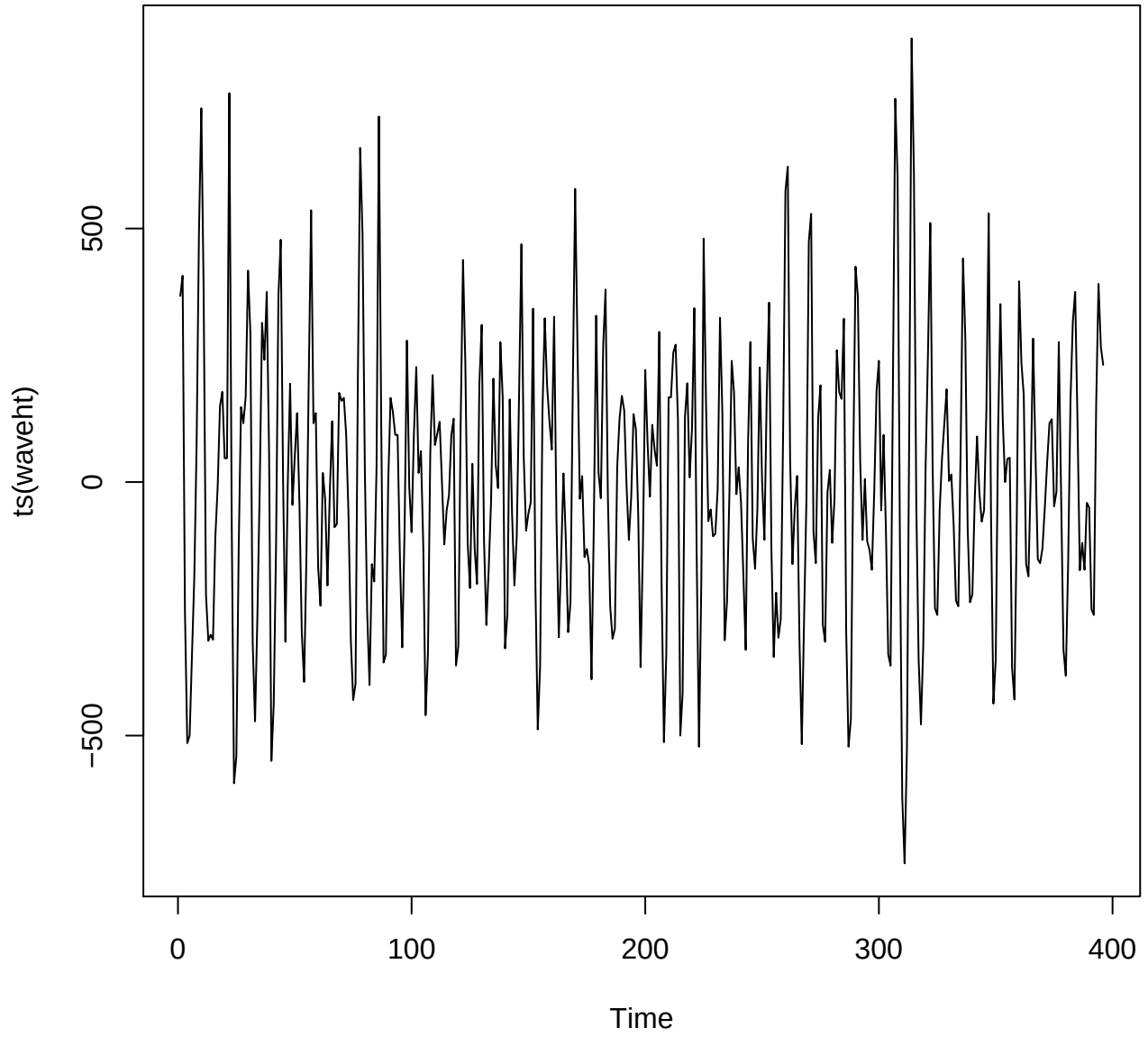
```
> wave.dat <- read.table("D:/MATH 5045/Data/wave.txt", header=T)
```

Note that depending on where you have stored the file, you will need to specify the correct path to it the read.table command. The data set already has its own header waveht, so header=T. We will do some plots to look at the data first.

```
> plot(ts(waveht))
```

produces the plot in Figure 1 over the entire 39.7 seconds. It is difficult to see whether there are any periods but it does look like there are no trends.

Figure 1: Plot of Wave Height over 39.7 seconds



We magnify the 1st 60 seconds in Figure 2.

```
> plot(ts(waveht[1:60]))
```

While there are ups and downs, these do not look like they are periodic. So we can assume that the time series is stationary.

The autocorrelations of a time series $x = (x_1, x_2, \dots, x_n)$ are stored in a vector `acf(x)$acf` in **R**. To see a chart of most of the acf the `acf` function in **R** plots the autocorrelation r_k versus k .

```
> acf(waveht)
```

See Figure 3. This chart is know as the *correlogram*.

In Figure 3, we can see that there is a positive correlation at say, lag 1, but negative at lag 2. The dotted line tells us which lags are significant at the 5% significance level. Anything positive above the dotted line is significant and anything negative below the negative dotted line is also significant. The dotted lines are drawn at

$$-\frac{1}{n} \pm \frac{2}{\sqrt{n}},$$

since under the null hypothesis that $\rho_k = 0$, it will be distributed approximately as normal $N\left(-\frac{1}{n}, \frac{1}{\sqrt{n}}\right)$.

The command

```
> acf(waveht)$acf
```

Figure 2: Plot of Wave Height over the 1st 60 seconds

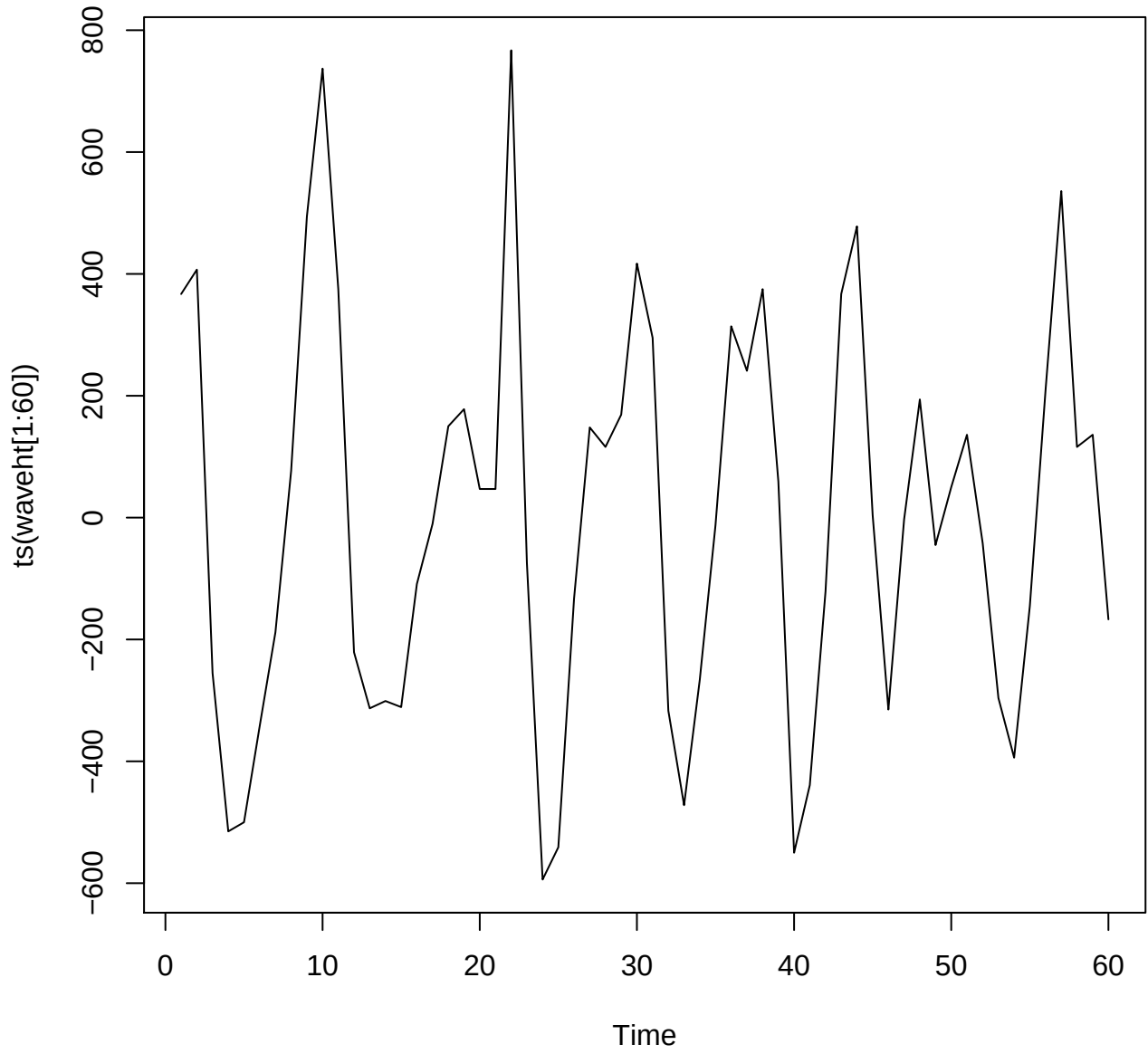
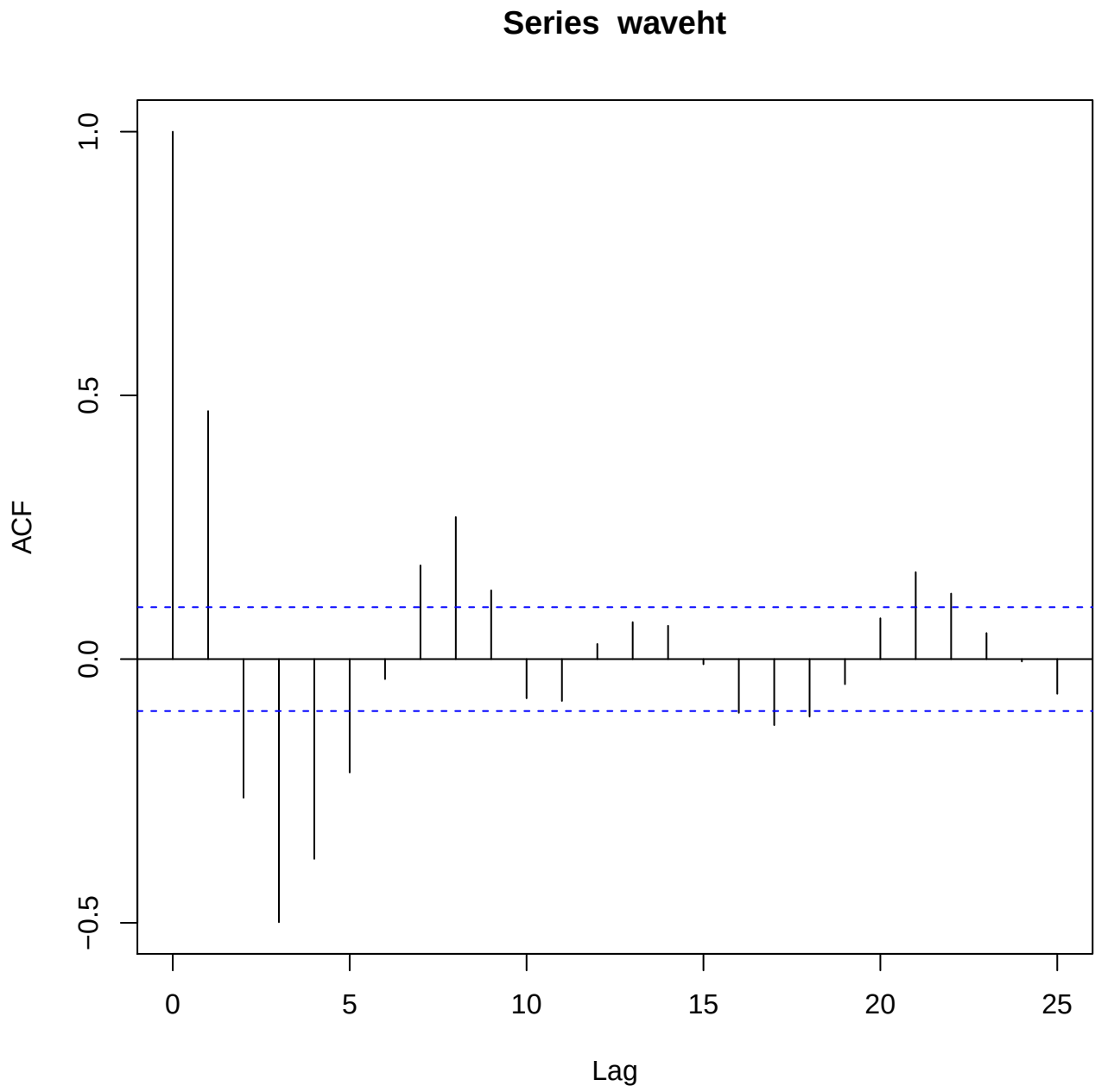


Figure 3: Autocorrelation of Wave Height Data

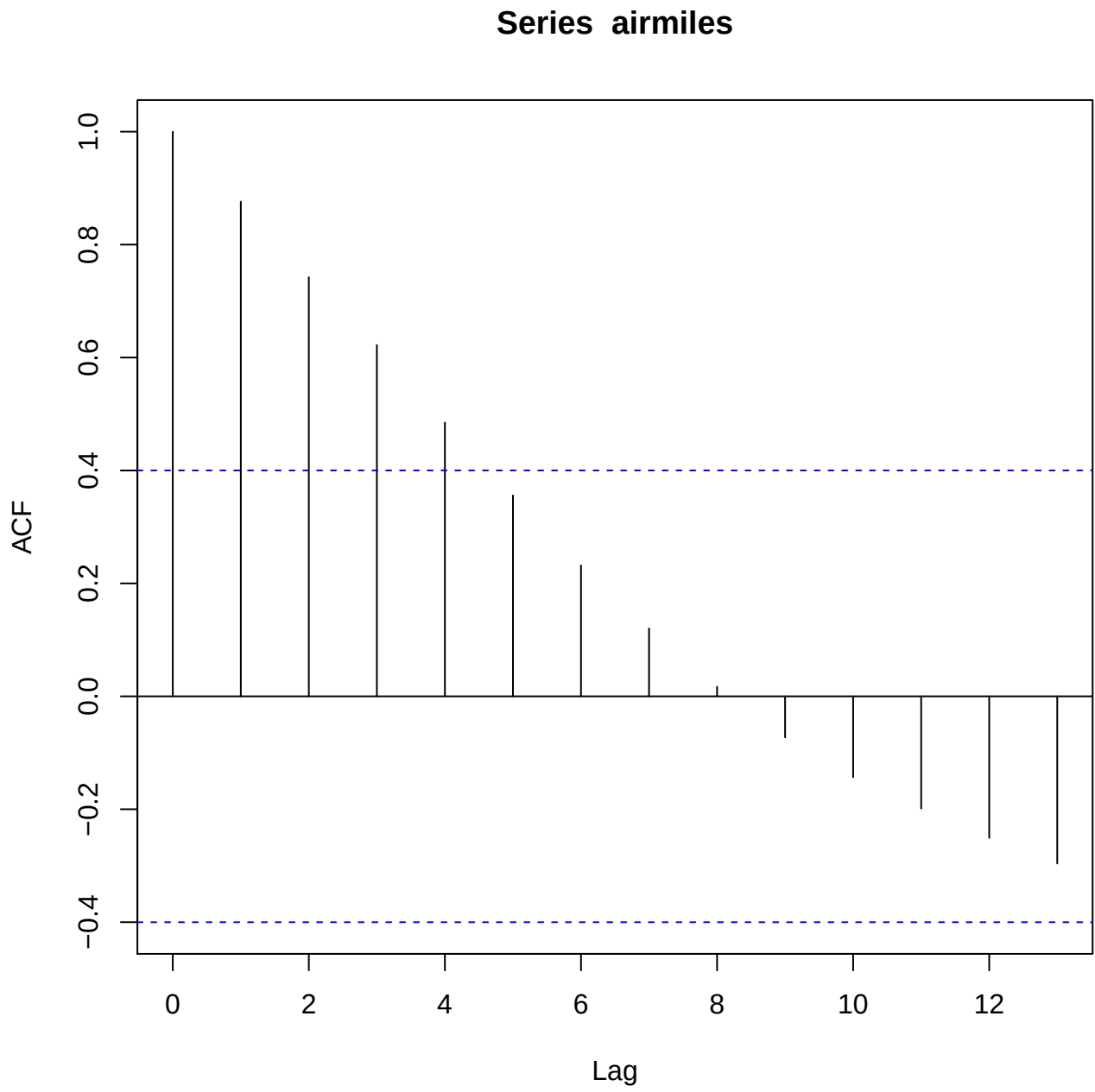


lists all the specific values of r_k . Note that r_k is given by the $[k+1,]$ row in the output in the **R** console. In order to get r_k alone, use `acf(waveht)$acf[k+1, ]`.

16. **Example 2.** Refer to the airmiles data in Chapter 1. This has an increasing trend. Even if this has a trend, we can still plot its correlogram. Using `acf(airmiles)`, we get the correlogram in Figure 4.

Note that if there is a trend, the correlogram also shows the trend.

Figure 4: Autocorrelation of Air Miles Data



17. **Example 3.** Refer to the Nottingham temperatures data in Chapter 1. This is periodic. The periodicity can be seen in its correlogram too in Figure 5 using `acf(nottem)`.

It shows that there is significant negative correlation between the current observation and those around half year earlier, and positive correlation between the current observation and those around one full year earlier.

18. Although the population acf is defined only for stationary series, the sample acf can be calculated for any series, including deterministic series and how the correlogram looks like can alert us whether there is a trend or seasonality.
- If you construct a time series that consists of a trend only, for example the integers from 1 to 1000, then the acf decreases slowly and almost linearly from one, see also Example 2 and Figure 4.
 - If you take a large number of cycles of a discrete sinusoidal wave of any amplitude and phase, the acf is a discrete cosine function of the same period.
 - If you construct a time series that consists of an arbitrary sequence of p numbers repeated many times, the correlogram has a dominant spike of almost 1 at p .

Tutorial 3.

1. Use the Nile river flow data, plot the correlogram for flows after 1899 and comment on its appearance.
2. Using some of the other data in **R** that you have identified as time series in Tutorial 1, plot their correlogram and comment on their appearance.

Figure 5: Autocorrelation of Nottingham Temperatures Data

